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1.6 Ethical, Legal, Cultural and Environmental Impacts

OCR GCSE Computer Science (J277) — Paper 1

Digital technology affects every aspect of modern society. Exam questions on this topic expect you to discuss **both sides** of issues (benefits AND drawbacks), reference **specific legislation by name and year**, and give considered, balanced arguments. This is one of the few areas where there are no definitively "right" answers — the marks come from the quality of your reasoning and the breadth of your discussion.

1.6.1 Ethical Issues

> **Ethics** in computing refers to the moral principles that guide how technology should be developed, deployed, and used. Ethical issues are not the same as legal issues — something can be legal but unethical (e.g., a company collecting excessive personal data within the law), or illegal but arguably ethical (e.g., a whistleblower exposing harmful practices).

Privacy and Surveillance

Mass data collection is now routine. Companies track browsing history, location, purchases, social media activity, and communications — often without users' full understanding or informed consent. Governments use surveillance cameras, facial recognition, and communications monitoring for law enforcement and national security.

- **Arguments for surveillance:** Helps prevent and solve crime and terrorism; protects national security; can improve public services (e.g., traffic management); assists in finding missing persons; deters antisocial behaviour.

- **Arguments against:** Infringes the right to privacy (a fundamental human right); creates a "chilling effect" on free speech (people self-censor if they know they are being watched); risk of misuse by authoritarian governments; disproportionately affects minority communities; data breaches can expose sensitive personal information; the power imbalance between those surveilling and those being surveilled.

AI and Algorithmic Bias

Artificial intelligence systems increasingly make or influence decisions about people — in hiring, criminal sentencing, credit scoring, healthcare, insurance, and content recommendation. These systems learn from historical data, which may contain biases.

- **Arguments for AI decision-making:** Can process more information than humans; more consistent (does not have bad days); faster; can identify patterns humans miss; potentially removes individual human prejudice.
- **Arguments against:** If trained on biased data, AI amplifies and systematises those biases at scale; "black box" decisions are difficult to explain, challenge, or appeal; accountability is unclear when an AI makes a harmful decision (who is responsible — the programmer, the company, the user?); can perpetuate discrimination against protected groups.

Digital Divide

The **digital divide** is the gap between those who have access to modern technology and internet connectivity and those who do not. This divide exists along several dimensions:

- **Economic:** Low-income households may not be able to afford devices, broadband, or data plans.
- **Geographic:** Rural areas often have slower or no broadband compared to urban areas.
- **Global:** Developing nations have far lower internet penetration than developed nations.
- **Age:** Older generations may lack digital skills or confidence with technology.
- **Impact:** Those without access are excluded from online education, banking, healthcare appointments, government services, job applications, and social connections. The COVID-19 pandemic highlighted how those without internet access were severely disadvantaged when schools and workplaces moved online.

Automation and Employment

Robots, AI, and software automation are replacing human workers across many industries — manufacturing, customer service (chatbots), transport (self-driving vehicles), retail (self-checkout), data analysis, and even creative fields.

- **Arguments for automation:** Increases efficiency and reduces costs; eliminates dangerous, repetitive, or unpleasant jobs; can work 24/7 without breaks; higher quality and consistency; frees humans for more creative, fulfilling work; new technology industries create new types of jobs.
- **Arguments against:** Mass job losses cause economic hardship, particularly for low-skilled workers; widening income inequality (benefits go to company owners, not displaced workers); ethical duty to retrain affected workers; transition period can be painful for communities; new jobs may require different skills that displaced workers lack; social and psychological impact of unemployment.

Cyberbullying and Online Harassment

Technology enables harassment at unprecedented scale. Online abuse can be anonymous, persistent (difficult to remove permanently), searchable, and reach a global audience. Particularly harmful to young people.

- Platforms have a responsibility to moderate content, provide reporting mechanisms, and protect vulnerable users.
- Victims may suffer serious mental health impacts including anxiety, depression, isolation, and self-harm.
- Schools, parents, and technology companies all share responsibility for education, prevention, and response.
- Legislation (such as the Online Safety Act 2023) places legal duties on platforms to protect users from harmful content.

Misinformation and Deepfakes

Social media algorithms are designed to maximise engagement, which means they often amplify sensational, controversial, or emotionally provocative content — including false information. Deepfake technology (AI-generated fake video/audio) makes it increasingly difficult to distinguish real from fabricated content.

- Misinformation undermines trust in institutions, science, and democratic processes.

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- Particularly dangerous during elections (propaganda), public health crises (anti-vaccine misinformation), and conflicts (disinformation).
- Solutions include fact-checking, media literacy education, platform moderation, and technical detection tools — but none is fully effective.

> **Exam Tip:** For ethical questions, always present **both sides** of the argument. Structure your answer: state the issue, give arguments for, give arguments against, and (if asked) give your reasoned conclusion. Do not just list points — explain and develop them. Use specific examples where possible. The best answers show awareness that these issues involve genuine tensions between competing values (privacy vs security, efficiency vs employment, innovation vs equality).

1.6.2 Key UK Legislation

You must know four key pieces of legislation by **name and year**, understand what each covers, and be able to apply them to specific scenarios.

Data Protection Act 2018 / UK GDPR

> The **Data Protection Act 2018** (incorporating the UK GDPR) controls the collection, storage, processing, and use of **personal data**. It replaced the 1998 Act and strengthened individuals' rights over their data.

Key principles — organisations must:

1. **Lawfulness, fairness, transparency** — process data lawfully and be transparent about how it is used.
2. **Purpose limitation** — collect data only for specified, explicit purposes and not use it for anything else.
3. **Data minimisation** — collect only the data that is necessary (do not collect data "just in case").
4. **Accuracy** — keep data accurate and up to date.
5. **Storage limitation** — do not keep data longer than necessary; delete it when no longer needed.

6. **Security** — store data securely using appropriate technical measures (encryption, access controls, secure backups).

Individual rights:

- **Right of access** — request a copy of all personal data held about you (Subject Access Request).
- **Right to rectification** — have inaccurate data corrected.
- **Right to erasure** — request deletion of your data ("right to be forgotten") in certain circumstances.
- **Right to object** — object to processing of your data for direct marketing.
- **Right to data portability** — receive your data in a machine-readable format to transfer to another service.

Enforced by the **ICO (Information Commissioner's Office)** with fines up to 17.5 million pounds or 4% of global annual turnover (whichever is greater).

Computer Misuse Act 1990

> The **Computer Misuse Act 1990** makes it a criminal offence to access or modify computer systems without authorisation. It was introduced in response to early hacking cases and has been amended several times to address evolving cyber threats.

Three key sections (frequently examined):

Section	Offence	Example	Maximum Penalty
Section 1	Unauthorised **access** to computer material	Guessing or cracking someone's password to access their email or social media account. Even just looking at data you are not authorised to see, without modifying anything, is an offence.	Up to 2 years imprisonment
Section 2	Unauthorised access with **intent to**	Hacking into a bank system with the intention of committing fraud; accessing a	Up to 5 years imprisonment

	commit a further offence	database to steal personal data for identity theft.	
Section 3	Unauthorised **modification** of computer material	Deploying malware (viruses, ransomware); deleting someone's files; launching a DDoS attack; modifying a website without permission (defacement).	Up to 10 years imprisonment

Section 3A (added 2006) makes it illegal to make, supply, or obtain tools intended for use in computer misuse offences.

Important distinction: Penetration testing (ethical hacking) is legal when performed with **written authorisation** from the system owner. Without authorisation, the same actions are criminal offences under this Act.

Copyright, Designs and Patents Act 1988

> The **Copyright, Designs and Patents Act 1988** protects **creative works** from being copied, distributed, modified, or sold without the creator's permission. Copyright is automatic — no registration is needed. The creator owns the copyright as soon as the work is created.

What is protected: Software source code and executables; music, audio recordings, and song lyrics; photographs and digital images; videos and films; written content (articles, books, blog posts, documentation); website designs; databases; game designs; artistic works.

Common violations in computing:

- Pirating software (downloading cracked versions or sharing licence keys)
- Downloading music, films, or TV shows illegally (torrenting copyrighted content)
- Copying code from another programmer without permission or proper licence
- Using images from the internet without the photographer's permission
- Copying and redistributing e-books or digital publications

Copyright lasts for the creator's lifetime plus 70 years. Penalties include fines and up to 10 years imprisonment for serious commercial piracy.

Freedom of Information Act 2000

Gives the public the right to request information held by **public authorities** (government departments, councils, NHS, police, schools, universities). Public authorities must respond within 20 working days. Exemptions exist for national security, personal data, law enforcement, and commercial sensitivity. Promotes government transparency and accountability. Enforced by the ICO.

1.6.3 Software Licences

Software licences define how software can be used, modified, and distributed. Understanding the difference between open source and proprietary software is essential.

Feature	Open Source	Proprietary (Closed Source)
Source code	Publicly available — anyone can view, study, learn from, and audit it	Kept secret — only the compiled executable is distributed. Users cannot see how it works internally.
Cost	Usually free to download, use, and distribute. Some companies offer paid support or premium features.	Usually requires purchase (one-time licence fee, subscription, or freemium model)
Modification	Users can modify and customise the software. Can adapt it for specific needs, fix bugs themselves, or add features.	Modification is not permitted. Users must use the software as-is and wait for the vendor to make changes.
Distribution	Can be freely redistributed (subject to licence terms like GPL, MIT, Apache). GPL requires that derivative works also be open source.	Cannot be copied, shared, or redistributed without the publisher's permission. Licence is typically per-user or per-device.
Support	Community-driven: forums, documentation, Stack Overflow, GitHub issues. Paid professional support available from specialist companies (e.g., Red Hat for Linux).	Professional vendor support: phone, email, live chat, guaranteed response times (SLAs). Training and certification programmes.

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Security	Transparent code means vulnerabilities can be found and fixed by anyone in the community — "many eyes make all bugs shallow." But attackers can also read the code to find exploits.	"Security through obscurity" — code is hidden. Vulnerabilities are found and fixed by the vendor on their schedule. Users depend entirely on the vendor to issue patches promptly.
Examples	Linux, Firefox, LibreOffice, Python, VLC, Android (AOSP), WordPress, Apache, MySQL	Microsoft Windows, Microsoft Office, Adobe Photoshop, macOS, iOS, Final Cut Pro

Types of Software by Availability

Type	Definition	Source Code?	Cost	Modify?	Example
Open Source	Source code is publicly available; anyone can view, modify, and distribute it (subject to licence terms)	Yes — publicly available	Usually free	Yes	Linux, Firefox, Python, LibreOffice, VLC
Proprietary (Closed Source)	Source code is kept secret; users receive only the compiled executable; terms of use set by the vendor	No	Usually paid (licence fee or subscription)	No	Microsoft Windows, Adobe Photoshop, macOS
Freeware	Software that is free to use but the source code is NOT available and modification is not	No	Free to download and use	No	Skype (older versions), older versions of Adobe Reader,

	permitted — "free as in beer, not free as in freedom"				WinRAR (technically)
Shareware	Software distributed free on a trial basis — either time-limited (expires after 30 days) or feature-limited (full features unlocked after payment)	No	Free trial; paid for full version	No	WinZip, many antivirus products, older games (shareware episodes)

Open Source Licences

Not all open source software is identical — the licence determines what you can and cannot do with it:

- **GPL (GNU General Public Licence)** — "copyleft" licence. You can use, modify, and distribute the software, but any derivative work must also be released under the GPL. Ensures open source remains open source.
- **MIT Licence** — very permissive. You can use, modify, and even include the code in proprietary (closed source) products with minimal conditions. Only requires attribution. Popular in commercial software development.
- **Apache 2.0** — permissive; similar to MIT but also grants an explicit patent licence. Allows use in proprietary products.

Creative Commons Licensing

Creative Commons (CC) licences are used for creative works (images, music, text, video) rather than software. They let creators specify how their work may be used. The licences combine four conditions:

Symbol	Code	Condition	Meaning

BY	Attribution	Required in all CC licences	You must credit the original creator when using or sharing their work
SA	ShareAlike	Optional add-on	If you create a derivative work, you must release it under the same CC licence (like GPL for creative works)
NC	NonCommercial	Optional add-on	You may not use the work for commercial purposes (selling it or using it to make money)
ND	NoDerivatives	Optional add-on	You may share the work but may not modify or create derivative works from it

Examples: **CC BY** = most permissive (credit only). **CC BY-NC-SA** = credit required, no commercial use, derivatives must use same licence. **CC BY-ND** = credit required, no modifications permitted.

> **Exam Tip:** Know the distinction: open source means source code is available and modifiable (under licence); freeware means free to use but source code is NOT available. A common exam trap is to confuse "free" (no cost) with "open source" (access to source code) — they are not the same thing. Always check what the question is actually asking: cost? Access to source code? Right to modify? Right to redistribute?

Recommending a Licence Type for a Scenario

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Scenario: A small charity with limited budget needs office software.
→ Open source (LibreOffice). Free to use, reducing costs. Community
  support is available. The charity does not need advanced features
  that justify the cost of Microsoft Office.

Scenario: A large bank needs a database system for financial records.
→ Proprietary (e.g., Oracle). Professional vendor support with
  guaranteed response times (SLA) is critical for a bank. The bank
  can afford the licence cost and needs the reliability, compliance
  features, and support guarantees that come with enterprise software.

Scenario: A student learning to program.
→ Open source (Python, VS Code). Free to use, widely documented,
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  large learning community. Being able to read source code aids
  learning. No cost barrier to getting started.

Scenario: A graphic design studio producing professional work.
→ Proprietary (Adobe Photoshop/Illustrator). Industry-standard
  tools with no open-source equivalent of the same quality.
  Client compatibility requires specific file formats. Professional
  support and training available.
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1.6.4 Cultural and Environmental Impacts

Cultural Impacts of Technology

Technology has fundamentally changed how people live, work, communicate, and consume media:

Area	Positive Impact	Negative Impact
Social media and communication	Connects people globally; enables communities around shared interests; amplifies marginalised voices; facilitates grassroots organising and activism	Spreads misinformation; creates echo chambers and political polarisation; can harm mental health (unrealistic comparisons, addiction, cyberbullying); reduces face-to-face social skills
Working practices	Remote working provides flexibility, eliminates commuting, opens opportunities regardless of location; video conferencing and cloud tools enable global collaboration	Blurs work-life boundaries ("always on" culture); can increase isolation; digital monitoring of employees raises privacy concerns; not all jobs can be done remotely (deepens inequality)

Entertainment and media	Streaming services provide vast libraries of content on demand; independent creators can reach global audiences; interactive entertainment (gaming) is now mainstream	Disrupted traditional industries (music, film, publishing, journalism); piracy challenges creators' livelihoods; concerns about screen time and addiction, especially for young people
Shopping and commerce	E-commerce offers convenience, wider choice, price comparison; enables small businesses to reach global markets	High street shops closing, changing town centres and communities; exploitation of gig economy workers (delivery drivers); concerns about monopoly power of large platforms
Education	Online learning increases access to education globally; resources available 24/7; personalised learning; enables lifelong learning	Digital divide excludes those without reliable internet or devices; concerns about quality of online-only learning; reduced social interaction between students
Health and wellbeing	Telehealth appointments increase access; health monitoring wearables; online mental health resources; AI-assisted diagnosis	Excessive screen time linked to health issues; social media linked to anxiety and depression in young people; technology addiction; reduced physical activity

Environmental Impacts

Negative Impacts	Positive Impacts
Data centres consume enormous amounts of electricity (estimated 1-2% of global electricity) for computing and cooling. Training large AI models has a significant and growing carbon footprint. Cloud computing shifts energy use from individual devices to centralised facilities, but overall energy demand is increasing.	**Smart grids** and IoT sensors optimise energy distribution, reduce waste, and help integrate renewable energy sources more effectively. Smart meters help consumers reduce energy usage.

E-waste (electronic waste) is one of the fastest-growing waste streams globally. Discarded electronics contain toxic materials (lead, mercury, cadmium, lithium, brominated flame retardants). Much e-waste is exported to developing countries where informal recycling exposes workers and communities to hazardous substances.	**Remote working** reduces commuting (fewer cars, less fuel), office heating/cooling, and business travel (replaced by video conferencing), significantly reducing carbon emissions.
Mining rare earth metals (cobalt, lithium, tantalum, neodymium) for device batteries, processors, and screens causes environmental destruction, water pollution, habitat loss, and human exploitation (including child labour in some cobalt mines).	**Precision agriculture** uses sensors, drones, GPS, and AI to optimise water usage, fertiliser application, and crop monitoring. Reduces waste, chemical runoff, and environmental impact while improving yields.
Planned obsolescence — manufacturers design devices with short lifespans, non-replaceable batteries, or stop providing software updates, forcing consumers to buy new devices more frequently. This creates unnecessary waste.	**Environmental monitoring** using satellite imagery, sensor networks, and AI tracks deforestation, pollution levels, wildlife populations, ocean temperatures, and climate change — providing data essential for conservation efforts.
Manufacturing and shipping — producing devices involves energy-intensive manufacturing processes and global supply chains that generate significant greenhouse gas emissions.	**Dematerialisation** — digital alternatives reduce physical resource use: e-books replace printed books, streaming replaces physical media (CDs/DVDs), digital tickets replace paper, video calls replace some travel, cloud storage reduces need for physical media.

Exam Tip: Know all **three sections of the Computer Misuse Act 1990** with specific examples for each — this is one of the most frequently examined topics in this unit. Be able to **recommend open source or proprietary software for a given scenario** with clear justification. For impact questions, **always give both sides** (positive AND negative) with specific examples. Name legislation by its **full name and year** — saying just "the Data

Protection Act" without the year loses marks. For ethical questions, structure your answer with clear paragraphs covering different perspectives.

Stakeholder Analysis

A **stakeholder** is any person or group who has an interest in, or is affected by, a technology, system, or policy. In computing exam questions, you are often asked to discuss how a technology or issue affects **different groups of people**. The key skill is showing that the *same technology* can have *very different impacts* on different stakeholders.

The main stakeholder groups you should be aware of for GCSE Computing:

- **Users** — individuals who directly use the technology (end-users, consumers, members of the public)
- **Developers / Programmers** — the people who design, build, and maintain the system
- **Businesses / Organisations** — companies that deploy or profit from the technology
- **Government / Regulators** — bodies that set laws and enforce rules (e.g., ICO, OFCOM, Parliament)
- **Society as a whole** — the broader public; communities; future generations

Stakeholder Analysis Example: Social Media Platform with Facial Recognition

Stakeholder	Potential Benefits	Potential Harms / Concerns
Users (individuals)	Convenient auto-tagging in photos; faster account recovery; accessibility features (e.g., describing photos for visually impaired users)	Privacy invasion — facial data collected without full informed consent; risk of data breach exposing biometric data; potential for misidentification; permanent biometric data cannot be changed if stolen
Developers	Opportunity to build innovative features; high demand for AI/ML expertise; commercial	Ethical responsibility for harmful uses of their technology; legal liability if system causes harm or breaches data protection

	opportunity; technical challenge and career development	law; pressure to build features management wants even if ethically questionable
Businesses (the platform)	Enhanced user experience drives engagement; richer data about users improves targeted advertising revenue; competitive advantage over rivals without the feature	Regulatory risk — GDPR/UK GDPR requires explicit consent for biometric data; risk of large fines from ICO; reputational damage if facial recognition misuse is publicised; cost of implementing system correctly
Government / Regulators	Potential law enforcement benefits (identifying criminals); ability to monitor use and regulate if harmful; opportunity to shape responsible AI policy	Must balance security benefits against civil liberties; enforcing compliance across global platforms is difficult; risk of authoritarian misuse of the technology if poorly regulated
Society	Can be used to find missing persons; increased security in some contexts; potential accessibility benefits at scale	Normalises mass surveillance; chilling effect on free expression (people behave differently if they know they are always identified); disproportionate false-positive rates for minority groups (algorithmic bias)

Exam Tip: Always name at least 3–4 stakeholders and explain how the SAME technology affects them *differently*. Do not just say "it is bad for everyone" or "it is good for everyone" — that misses the point. The highest-mark answers acknowledge genuine trade-offs: a feature that benefits a business (increased engagement, ad revenue) may simultaneously harm the user (privacy loss) and concern regulators (compliance risk). Show you understand that technology creates winners and losers.

Stakeholder Analysis Example: Automation in a Supermarket (Self-Checkout)

Stakeholder	Impact

Customers (users)	Faster checkout for small purchases; available 24/7; some find it difficult to use (older customers, those with disabilities, those with large trolleys)
Checkout workers	Job losses and reduced hours as machines replace roles; remaining workers may be redeployed to assist self-checkout bays, but often on lower wages; deskilling of workforce
Supermarket (business)	Reduced long-term labour costs; higher throughput during peak times; significant upfront capital investment in machines; liability for errors
Government	Reduced income tax and National Insurance revenue from lost jobs; potential increased social security spending; incentive to retrain displaced workers
Society	Reduced employment opportunities in retail (a common entry-level sector); town centre decline if combined with online shopping growth; concern about long-term employment levels

Cultural Issues — Digital Divide and Automation

The Digital Divide — In Depth

> The **digital divide** is the gap between those who have meaningful access to digital technology (devices, reliable internet, digital literacy) and those who do not. It is not a single divide but a series of overlapping inequalities affecting different groups in different ways.

The digital divide has multiple dimensions:

Dimension	Who is Affected	Specific Impact
Economic	Low-income households, people in poverty	Cannot afford devices (a smartphone or laptop can cost hundreds of pounds), broadband subscriptions (£30–£60/month), or mobile data plans. Children in these households struggled most when schools moved online during COVID-19 lockdowns.

Geographic	People in rural and remote areas	Broadband infrastructure (especially fibre) has been built in profitable urban areas first. Rural areas often have slow ADSL connections or rely on expensive satellite broadband. Poor mobile signal further limits access. This affects access to online services, remote working, and telehealth.
Age	Older adults, elderly people	Many older people lack digital literacy — they did not grow up with computers and may find technology confusing, intimidating, or inaccessible. Loss of physical bank branches and government services moving online can exclude them from essential services.
Disability	People with visual, hearing, motor, or cognitive impairments	Poorly designed websites and apps create barriers for users who rely on screen readers, voice control, or alternative input devices. WCAG (Web Content Accessibility Guidelines) set standards for accessible web design — but many sites do not comply.
Language	Non-English speakers	Much of the internet's most valuable content (educational resources, technical documentation, business tools) is in English. Non-English speakers may find fewer resources, less support, and greater difficulty navigating online systems.
Global (North-South divide)	People in developing nations	Internet penetration in some sub-Saharan African nations remains below 30%, compared to 90%+ in Western Europe. Lack of infrastructure, high device costs relative to incomes, and limited electricity access all contribute.

Consequences of the Digital Divide

- **Exclusion from services** — banking, government benefits, job applications, healthcare appointments, and education are increasingly online-only. Those without access cannot participate on equal terms.

- **Widening inequality** — those with good digital access can upskill, work remotely, access better prices, and participate in the digital economy. Those without are left further behind, deepening existing social and economic inequality.
- **Educational disadvantage** — students without home internet or devices cannot complete homework, access online resources, or participate in remote learning, affecting academic outcomes.
- **Social isolation** — particularly for elderly people, lack of digital access cuts them off from communication (video calls with family), community resources, and social support networks that have moved online.

Mitigations — Addressing the Digital Divide

- **Government broadband investment** — the UK Government's Project Gigabit aims to bring gigabit-capable broadband to at least 85% of the UK by 2025, prioritising rural areas that the market will not reach commercially.
- **Digital skills training** — free courses (e.g., delivered by libraries, adult education centres, charities like Age UK's digital inclusion programmes) help older and less confident users develop essential digital skills.
- **Device loan schemes** — some schools, councils, and charities loan devices to students and families who cannot afford their own.
- **Accessibility standards** — WCAG (Web Content Accessibility Guidelines) provide a framework for making websites usable by people with disabilities. Public sector websites in the UK are legally required to meet WCAG 2.1 AA standard.
- **Low-cost tariffs** — some ISPs offer social tariffs (discounted broadband for households receiving certain benefits), reducing the cost barrier.

Impact of Automation and AI on Employment

Advances in robotics, AI, and software are transforming the labour market. The impact is uneven across sectors and skill levels:

Aspect	Detail
Jobs at highest risk of	Routine, repetitive, and rule-based tasks are most easily automated. Examples: manufacturing assembly (industrial robots), data entry, basic customer service

automation	(chatbots), toll booth operation, supermarket checkout, some aspects of accounting and legal research, radiological image analysis.
New jobs created	New technology industries create roles that did not previously exist: AI/ML engineers, data scientists, cybersecurity analysts, cloud infrastructure engineers, drone operators, electric vehicle technicians, social media managers, app developers, robot maintenance technicians. Historically, technology has created more jobs than it has destroyed — but the transition is difficult.
Retraining challenges	Workers displaced from routine jobs often lack the qualifications and skills for new technical roles. Retraining takes time and money. Older workers may struggle to retrain for very different careers. The burden of retraining typically falls on individuals rather than employers or the state.
Income inequality	Benefits of automation (lower costs, higher productivity) accrue mainly to business owners and shareholders rather than workers. Workers in automated sectors face wage stagnation or job loss while profits increase. This can widen the gap between rich and poor.
Changing nature of work	Growth of the gig economy (Uber, Deliveroo, freelance platforms) — flexible but insecure work without employment rights, pensions, or sick pay. Automation may simultaneously create gig-economy demand while destroying stable employment.

Balanced perspective: Automation has historically raised living standards overall — industrial automation created prosperity that manual labour could not have achieved. But the transition period causes real hardship for individuals and communities. The question is not whether automation is good or bad, but how the benefits are distributed and how displaced workers are supported through the transition.

> **Exam Tip:** For digital divide questions, always address at least three different types of divide (economic, geographic, age are the safest three) with specific consequences for each. Do not just list the types — explain the real-world impact (e.g., "elderly people without internet access cannot book GP appointments online, which has moved away from phone booking in many practices"). For automation and employment, always present both sides: efficiency and new job

creation on one side; job losses, retraining costs, and inequality on the other. The strongest answers acknowledge that both are true simultaneously.